

Tanvir Mahmud Mahim

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🎓 EDUCATION

Dec 2022	B.Sc. in Electrical and Electronic Engineering (EEE), BRAC University
Jan 2019	- CGPA: 3.77/4.00 Average Marks: 86.20% Academic Honors: High Distinction - Certification: World Education Services (WES) Certified, Equivalent to a US Bachelor's Degree - Supervisor: Dr. A.H.M.A. Rahim [Retired (Dec 2024)] - Thesis: BRACU-DSpace-2023

🎓 HONORS AND AWARDS

Quality Journal Publication Award for Scopus-indexed Q1 journal publications (2025, 2026)

VC's and Dean's List Award for academic excellence at undergraduate study (2020-2022)

University Merit Scholarship for academic excellence at undergraduate study (2020-2022)

💡 RESEARCH INTERESTS

- Quantum Optics and Photonics
- 2D Quantum Materials
- Wide Bandgap (WBG) Semiconductor Devices
- Quantum Physics
- Electronic Design Automation (EDA)
- Machine Learning (ML)

🔬 RESEARCH EXPERIENCE

Present Jan 2025	Squeezed-Light Microcombs, Diamond Quantum Sensing and Photonic ML Supervisor: Dr. A.S.M. Mohsin, BRACU, EEE • Developed a numerical model of a photonic-molecule 4H-SiC soliton-crystal microcomb enabling squeezing extraction beyond 3 dB limit [Code] [Optics Express 2026] • Theoretically and numerically investigated a pick-and-place diamond micromembrane NV quantum sensor for imaging single vortices and quantifying vortex-induced losses in tantalum superconducting circuits [Code] [under review] • Developed closed-loop ML frameworks for quantum photonic instrumentation and photonic neural inference, in collaboration with the University of Memphis, USA: Closed-loop, event-driven machine learning for autonomous triage of single-photon emitters [Code] [under review]; Photon-efficient neural inference over optical links [Code] [under review] • Developed learned generative process digital twins for yield-aware inverse design of photonic sensor front-ends [Code] [under review]
July 2026 July 2024	GaN & 2D Semiconductor Devices and Physics-Guided Design Supervisor: Dr. M. Mosaddequr Rahman, BRACU, EEE • Resolved the conflicting hole effective masses of the polarization-induced GaN/AlN 2D hole gas through a parameter-free, self-consistent six-band envelope-function analysis [Code] [under review] • Developed a TCAD-based model relating two Raman phonon signatures to fixed edge charge in monolayer semiconductor nanoribbon transistors. [Code] [under review] • Investigated the combination of strain and ferroelectricity in WSe₂ transistors for fast, low-power non-volatile logic [Code] [under review] • Developed a fabrication-aware, physics-informed adjoint framework with RL for inverse design of CMUTs and photonic sensors. [Code], [under review], [IEEE Sensors Journal 2025] • Developed physics-guided circuit design frameworks, in collaboration with Dr. Nadim Chowdhury, BUET: adjoint co-optimization of a monolithic E-mode GaN-on-SOI fractional-N PLL with a HEMT varactor [Code] [under review]; graph RL with an adaptive fuzzy reward for transferable analog transistor sizing, with GlobalFoundries, Inc., USA [Code] [under review]
June 2024 May 2023	Learning-Based Grid Control and Bifacial Photovoltaics Supervisor: Dr. A.H.M.A. Rahim, BRACU, EEE [Retired (Dec 2024)] • Investigated multi-machine network control with RL-based fuzzy inference [JESTECH (Elsevier) 2026] • Developed an RL-driven fuzzy-attention inference controller for microgrids. [IEEE TFS 2025] • Developed a novel bifacial module [IEEE TENSYP 2023], [IEEE JPV 2024]

SELECTED PUBLICATIONS (FIND ALL HERE:

- 2026 **T. M. Mahim**, M. M. Rahman and A. S. M. Mohsin, “Overcoming the 3 dB Squeezing Extraction Limit in Silicon Carbide Microcombs with a Photonic Molecule”, *Optics Express*, 34(18), 34822 (2026). [\[Paper\]](#)
- 2026 **T. M. Mahim**, M. M. Rahman and A. S. N. Huda, “Adaptive Takagi-Sugeno-Kang Fuzzy Inference with Reinforcement Learning to Withstand Extreme Low Voltages of DFIG-WT”, *Engineering Science and Technology, an International Journal*, 80, 102435 (2026). [\[Paper\]](#)
- 2025 **T. M. Mahim**, M. M. Rahman and A. H. M. A. Rahim, “Hierarchical Inverse Design Framework for Unit-Cell CMUTs With Attentive Gated Recurrent And Fully Connected Dense Layers”, *IEEE Sensors Journal*, 25(13), 26104-26116 (2025). [\[Paper\]](#)
- 2025 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, “Adaptive Fuzzy Attention Inference to Control A Microgrid Under Extreme Fault On Grid Bus”, *IEEE Transactions on Fuzzy Systems*, 33(6), 1815-1824 (2025). [\[Paper\]](#)
- 2024 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, “Weather Responsive Multidimensional Photovoltaic Efficiency Model for Simulation of Custom-Built Bifacial Panel”, *IEEE Journal of Photovoltaics*, 14(5), 848-860 (2024). [\[Paper\]](#)
- 2024 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, “Review of Mono- and Bifacial Photovoltaic Technologies: A Comparative Study,” *IEEE Journal of Photovoltaics*, 14(3), 375-396 (2024). [\[Paper\]](#)

OPEN-SOURCE RESEARCH SOFTWARE (TAN-MM-ORG · PYPI)

ramansep	Strain and carrier-density separation in Raman maps of 2D materials; linear inversion with uncertainties
kpenvelope	Six-band k-p envelope-function solver for wurtzite heterostructures, self-consistent with Poisson
sqzcomb	Detected-squeezing spectra of Kerr microcombs; photonic molecule surpasses the single-ring 3 dB limit
absnoise	Andreev-bound-state occupation noise and detector budgets for proximity Josephson junctions
sparq-triage	Event-driven reinforcement-learning triage of single-photon emitters against exact photon statistics
cavsqueeze	Beyond-mean-field cavity-mediated spin squeezing of solid-state clock-transition ensembles

TECHNICAL SKILLS

TCAD & Device Simulation	DEVSIM · Synopsys Sentaurus TCAD
Multiphysics & Electromagnetics	COMSOL Multiphysics · Ansys HFSS
First-Principles & Atomistic	Density functional theory (Quantum ESPRESSO) · Wannierization (Wannier90) · Molecular dynamics (LAMMPS) · Micromagnetics (MuMax ³)
Circuit, Layout & EDA	Cadence Virtuoso · Keysight ADS · Xyce · ngspice · OpenVAF · GDSFactory
Machine Learning	PyTorch · JAX, Optax, Flax · TensorFlow & Keras · scikit-learn

PROFESSIONAL EXPERIENCE

Present June 2025	Adjunct Lecturer, Department of EEE, BRACU Courses: Solid-State Devices, Numerical Methods and Microprocessor & Embedded Systems
Present June 2023	Research Assistant, Department of EEE, BRACU Collaborated with industries and international universities; assisted undergraduate capstone projects
June 2026 Dec 2025	Consultant, GIXSystems.ai, USA Technology development of a 100 V GaN-on-Si power HEMT discrete device (tape-out with X-FAB, Erfurt)
May 2023 Jan 2022	Teaching Assistant, Department of EEE, BRACU Courses: DC & Magnetic Circuits, Energy Conversion, Signals & Systems and Control Systems

SYNERGISTIC AND LEADERSHIP ACTIVITIES

Present Jan 2021	Student Member IEEE · Electron Devices Society (EDS) Power and Energy Society (PES)
Nov 2023 Feb 2023	Aspire Institute Harvard University, USA · Aspire Leaders Program 2023

REFEREES

Dr. Md. Mosaddequr Rahman · Professor & Chairperson, Dept. of EEE, BRAC University, Bangladesh · mosaddeq@bracu.ac.bd

Dr. Abu S. M. Mohsin · Associate Professor, Department of EEE, BRAC University, Bangladesh · asm.mohsin@bracu.ac.bd

Dr. A.S. Nazmul Huda · Associate Professor, Department of EEE, BRAC University, Bangladesh · nazmul.huda@bracu.ac.bd

Dr. Nadim Chowdhury · Associate Professor, Department of EEE, BUET, Bangladesh · nadim@eee.buet.ac.bd

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